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PART TWO

# *Kinship Dreams*

CLASSIFYING

PLANT SYSTEMATICS



Give name to the nameless so it can be thought.  
The farthest horizons of our hopes and fears are cobbled  
by our poems, carved from the rock experiences of our daily lives.

AUDRE LORDE | *Sister Outsider: Essays and Speeches*



# The Categorical Impurative

## *Names, Norms, Normings*



Do you know that even when you look at a tree and say, “That is an oak tree,” or “that is a banyan tree,” the naming of the tree, which is botanical knowledge, has so conditioned your mind that the word comes between you and actually seeing the tree?

To come in contact with the tree you have to put your hand on it and the word will not help you to touch it.

JIDDU KRISHNAMURTI | *Freedom from the Known*

Purism is a de-collectivizing, de-mobilizing, paradoxical politics of despair. This world deserves better.

ALEXIS SHOTWELL | *Against Purity*

## Prologue

As a South Asian immigrant in the United States, I often meet my second-generation counterparts. When I first came to this country and asked their names, I was amused, and sometimes bristled, when they slightly or completely “mispronounced” their names. While I thankfully suppressed my impulse to “correct” them, it was only well into my forays into ethnic studies that I came to question the idea of “mispronunciation.” I had to ask myself, Who made me the judge? As I thought about my travels around India, I remembered how the same name is pronounced differently in different regions of the subcontinent. Indeed, on many an occasion, people from various states from northern India corrected me when I proclaimed my name, written and spoken, as Banu instead of Bhanu. This is a Sanskrit name, they told me. Maybe so, I learned

to respond, but I'm Tamilian. Tamil does not have the letter or sound "bha." This is how my parents spelled my name on my birth certificate. I'm claiming it! With the rising Hindi chauvinism in India, I've dug even deeper in claiming the Banu—it may be a Sanskrit word, but this is how I pronounce it. Yet many (including some in my family) still insist on writing to me as and calling me Bhanu. Names are personal, I have come to recognize. I ought to pronounce a name as given and requested by the bearer of the name, not through sanctified politics of purity. The "correct" name is always shrouded in structures of power—questions about authenticity, about right and might. As Max Weinreich quips, "a language is a dialect with an army and navy."<sup>1</sup> More importantly, in formerly colonized countries like India, where English remains one of the official languages, local languages are transliterated into the English alphabet. You could be Rina or Reena, Sita or Sitha, or Seeta or Seetha. To complicate matters, with the rise of Hindu nationalism, India in recent years has seen a huge uptick in numerology, leading to exuberant additions and omissions of vowels and consonants to take numerological considerations into account. Just consider Bollywood actors: Devgan became Devgn, Yash became Yassh, Khurana became Khurrana, Irfan became Irrfan, and Sanjana became Sanjjanaa.<sup>2</sup> With nationalist fervor, cities have embraced renaming. Did I grow up in Calcutta, Bombay, Madras, and Calicut or in Kolkata, Mumbai, Chennai, and Kozhikode?

My experiences in India and then an immigrant in the west are shaped by histories of hybridities. I have developed a flexibility and openness to names and naming and have come to terms with the potent politics behind multiple spellings, naming, renaming, and ever-changing pronunciations. Naming is a political act, and renaming likewise is not only symbolic but a kind of reckoning with the past. I'm also rather unmoved by claims of the sanctity of origin stories, of properness, of the immutability of tradition or culture. To me, words evolve, contexts change, and the world moves on. Today when I meet a second-generation Indian immigrant, I take their name as a reflection of their history and not a travesty of language.

## Bringing Order to Plant Worlds

My experiences with names and naming shape my analyses and reflections on botany. Botanical naming and nomenclature are deeply intertwined with how plants are organized into classificatory systems (fields of taxonomy and systematics). While in the sciences, papers published five years ago are considered “old,” nomenclature is one subfield that remains rooted in history.<sup>3</sup> Naming and taxonomy have ongoing and interrelated histories, and I explore them both in this chapter.

Most of the vast literature in the field of taxonomy describes the goal as bringing “order” to the diversity of nature. I’ve been struck by how often I’ve seen some version of the following sentiments:

Man is by nature a classifying animal. His continued existence depends on his ability to recognize similarities and differences between objects and events in his physical universe and to make known these similarities and differences linguistically. Indeed, the very development of the human mind seems to have been closely related to the perception of discontinuities in nature.<sup>4</sup>

As the story goes, because we are a classifying species, people across the globe brought “order” to the natural world by naming local plants and animals. But what is order? A group that categorizes the world into three categories—edible, inedible, and poisonous—has introduced an important practical and life-sustaining order. There is order everywhere, but the “order” brought about by botany is different and is deeply implicated in the histories of colonialism. As western explorers traversed the globe, they encountered diverse sets of nomenclatures and “order.” Prescientific societies (including in the west) also classified, named, and identified living things. In botanical lingo, these are labeled “folk taxonomies,” a patronizing and marginalizing term.<sup>5</sup> The trouble with folk taxonomies, the argument goes, is that “these names vary in a chaotic fashion from place to place and time to time.” So early in Western history arose “a professional group whose duty it was to bring order to this chaos.”<sup>6</sup> This group, systematists or taxonomists, brought botanical order. Their subdiscipline attempted to classify all living organisms into a universal, single, and stable system. “Lumpers” lump organisms by similarities and “splitters” note

differences as the grounds for classification. Tensions remain between lumpers, who define taxonomic groups broadly (fewer groups but larger numbers within each), and splitters, who multiply the numbers of groups and names (many groups with small and precise composition).<sup>7</sup>

Why do we need names, nomenclatures, and classifications? It is useful to begin with some definitions. Scientific names are based on a Linnaean binomial system. Plant naming and nomenclatures are regulated by a community of taxonomists—particularly those serving on the committee to create the *International Code of Nomenclature for Algae, Fungi, and Plants*. The members, mostly western scientists, meet regularly and set guidelines and rules for an active and ongoing process of nomenclature. Taxonomy is the theory and practice of identifying, describing, naming, and classifying organisms—that is, identification of taxa, or scientifically codified groups. While the terms *taxonomy* and *systematics* are often used interchangeably, Ernst Mayr defines *taxonomy* as “the theory and practice of classifying” and *systematics* as “the study of the diversity of organisms,” grounded in describing evolutionary relationships.<sup>8</sup> Both taxonomy and systematics use morphological, behavioral, genetic, and biochemical observations to classify organisms.<sup>9</sup>

Why do we need a singular universal system? And if we do need one, why did the colonial west get to determine it? Why is the scientific west never a “folk” language? The very language I’m forced to use belies the issue. What is the problem if people around the world develop their own locally coherent folk taxonomies as science? Why can’t we have a diversity of botanical languages, like we have a diversity of linguistic ones? Who needs universal stories? Is the scientific story truly a universal and coherent one? Is it “superior” to folk taxonomies? If so, how and for whom? Who gets to tell the scientific story? In what language is it told? Is there consensus about this story and its attendant systems? As you will guess, at the heart of the story are colonialism and coloniality—colonial exploration, colonial expansion, colonial profits, and colonial rule—all of which shaped the field of botany and the modern plant sciences, especially the naming, nomenclature, and classification of plants. Even in postcolonial nations (as we see in the next chapter), botanical infrastructures remain mired in western and colonial logics.

## A Brief History of Plant Nomenclature, Taxonomy, and Systematics

I begin with a brief history to show how colonial ideas shaped modern botany. This chapter is organized more or less chronologically: “Folk Taxonomies and the Prehistory of Plant Naming,” “Linnaeus and the Legacy of Binomial Nomenclature,” and “The Rise of Evolutionary Sciences and the Advent of DNA.” In each section, I first narrate the history of the field and then reflect on its colonial afterlives. I follow with scientific infrastructures of naming and their salience and shortcomings. Finally, I conclude with a discussion of the “tree of life.” What began as a study of the morphology of plants has increasingly become a study of the molecular. I end with a discussion of some key debates and themes in the history of plant taxonomy and reflections about botanical decolonizing.

### FOLK TAXONOMIES AND THE PREHISTORY OF PLANT NAMING

The history of botany begins with early and diverse understandings of plants across the world thousands of years ago (deemed folk knowledges). The roots of scientific botany, however, are usually traced to the Greeks and Romans. Genealogies acknowledge a long list of individuals who attempted to order plant worlds; they are exclusively men and from the west. Once plants were named, scholars attempted to create a taxonomy or system to organize them, looking for similarities and differences. Some of the key figures include Theophrastus (370–285 BC), often referred to as the “father” of botany, who classified plants by their form, as herbs, shrubs, trees, and so on; Andrea Caesalpino (1519–1603); Gaspard Bauhin (1560–1624); John Ray (1628–1705); and Joseph Pitton de Tournefort (1656–1708).<sup>10</sup> The early history of botany was attached to medicine, focused on medicinal herbs. Indeed, most great botanists from the sixteenth to eighteenth centuries, including Linnaeus, were professors of medicine or practicing physicians.<sup>11</sup> Early botany in the sixteenth century linked botanical medical treatments with plant habitats. For example, God put willows in wet habitats because men in wet habitats, prone to rheumatism, could be served by the willow.<sup>12</sup> A closer look at the scientific history of taxonomy

therefore betrays its own western “folk” roots since so much of plant understanding in the natural world relied on anthropocentric elements.<sup>13</sup>

Before the seventeenth century, homological thinking (prevalent during the Renaissance) was shaped by observations of correspondences and similarities among living things.<sup>14</sup> Botanical narratives reveled in the deep and rich connections among life on earth. Yet pre-Linnaean and pretheoretical taxonomies did not reduce nature into one “system.”<sup>15</sup> Rather, these were embrangled worlds structured by a vertical and hierarchical principle, where “society had a ‘natural’ order and nature had a ‘social’ order.”<sup>16</sup> For example, the *Doctrine of Signatures* attempted, prior to the end of the sixteenth century, to predict effective herbal remedies for various human illnesses.<sup>17</sup>

Michel Foucault traces the epistemological shifts from this earlier order into one marked by rational taxonomies and careful attention to the kind and categories of organisms—human, animal, plant—and eventually divisions between them. Foucault argues that this modern epistemology separated word and language from signs and forms of truth, thus dividing history from science.<sup>18</sup> While “premoderns” found endless, rich, and enchanted connections between nature and culture, modernity ushered in an absolute dichotomy between nature and culture.<sup>19</sup>

Folk taxonomies served botanists well until the great Age of Exploration (a euphemism for colonialism if I ever heard one!). Plant hunting in the late eighteenth century and most of the nineteenth unleashed a flood of “new” species that Europeans had never encountered before. For hundreds of years, Endersby argues, European naturalists believed that the lists of plants and animals they had inherited, especially from the Greeks, were largely complete.<sup>20</sup> When colonists discovered a treasure trove of new plants, animals, and peoples that they had no names for, they gave them names. At first Europeans gave these organisms new names, often wildly inaccurate translations and transcriptions of indigenous names or very long descriptive names. Many amusing stories are chronicled in the literature. For example, Dutch explorers of the Americas listed a long Arawak plant name that later turned out to mean, “I don’t know this one so I’ll have to ask my uncle.”<sup>21</sup> Or take the Latin name for one particular plant species: *Acaciae quadammodo accedens, Myrobalano chebulo Veslingii similis arbor Americana spinosa, foliis ceratoniae in pediculo geminatis, siliqua bivalve compressa corniculate seu cochlearum vel arietino-*

*rum cornuum in modum incurvate, sive Unguis cati* (“A spiny American tree, in some ways resembling *Acacia*, similar to Vesling’s *Myrobalanus chebulae*, with *Ceratonia* leaves in pairs at the pedicle, a silique with two valves, which is compressed, and horn-shaped or curved like the horns of snailshells or rams’ horns, or like a cat’s claws”).<sup>22</sup> As Tod Stuessy argues, “Just about everything under the Sun was used.”<sup>23</sup> These is why the simplicity of Linnaeus’s system emerged as so useful.

The colonists rapidly named their “discoveries” in the colonized worlds, creating a virtual “new Babel.”<sup>24</sup> But the problem for colonists was that the same plant or animal had different names in different parts of the colonies. How to tell if the aromatic spice in one region was the same as the one in another? By the eighteenth century, the colonists felt the urgent need for a consistent system. After all, how could they exploit natural resources efficiently if they did not know what plants existed where? Might they be able to grow these in their homelands? This created the grounds for a botanical nomenclature system with standardized names, well-developed life history details, good ecological plant characterizations, and a standardized hierarchical system of organization. Eventually a scientific infrastructure of botanical nomenclature emerged—standardized diagrams, herbarium sheets that displayed the plant with strict protocols, and key biological and contextual knowledge about the plant. The herbarium sheets traveled to find homes in colonial institutions, where they often remain to this day.

Much of colonial exploration was about utility. Ernst Mayr argues that when botany moved from its roots in medicine to become an independent science, and with the great colonial expeditions and the burgeoning of new species, the first concern was bringing order; eighteenth- and nineteenth-century botany was largely taxonomy.<sup>25</sup> Early taxonomy focused on classification through morphological similarities and differences. This transformed into plant systematics with its focus on evolutionary relationships, the topic of the next section.<sup>26</sup> Later, molecular technologies further transformed the field.

**COLONIAL ROOTS AND AFTERLIVES** Several aspects are worth noting in botany’s early historical narration. First, just as Columbus “discovered” America, so did white colonists “discover” plant species in the colonies. Species widely used for centuries by local peoples emerge in western narratives

as ones of “discovery,” and the discoverer’s role is forever enshrined in the scientific names of the species.

The vast apparatus of knowledge systems of indigenous medicine become fodder for western innovation. Hypocrisies abound. Indigenous systems and peoples are deemed prescientific, barbaric, and lacking civilizational value, yet their vast knowledge systems and natural resources were (and are) extracted by colonists.<sup>27</sup> Colonial bioprospecting is today’s biopiracy.<sup>28</sup> A western-dominated botanical establishment and infrastructure continues to set the terms of the debate.

Colonial logics enshrined plant official names in Latin (or latinized terms). With the rise of climate change and the continued onslaught against forests of the world, there is an obsessive anxiety about the loss of unnamed species. In 2015 alone, about 2,034 new species were discovered.<sup>29</sup> Some botanists estimate that of over 8.7 million terrestrial species, only about 14 percent are presently known to science (not counting the diverse world of protists, fungi, bacteria, and viruses).<sup>30</sup> In short, we need to speed up our taxonomic record and the “discovery” of new species to better chronicle and mourn the species we are losing!<sup>31</sup> A profound irony. This obsession with counting species (producing data) betrays an enduring logic of coloniality: a study of plants without people.

#### LINNAEUS AND THE LEGACY OF BINOMIAL NOMENCLATURE

In the history of botany, and indeed biology, the name of Linnaeus looms large. Undoubtedly one of the most celebrated and influential figures, Carolus Linnaeus (1707-1778) is regarded by disciplinary historians as the “father” of systematics. Linnaeus is important for two reasons—he created the first system of binomial nomenclature of plants that endures to this day, and he produced a classification and organizational system for plants. He also created a hierarchical model that followed Aristotelian representations of nature in three kingdoms—minerals, plants, and animals. In 1735, he published the *Systema Naturae*, where he organized biological life into a hierarchical set of groups—species into genera, genera into orders, orders into classes. His system was simple, a sexual system of classification based on the number and position of

the reproductive organs in a flower.<sup>32</sup> Linnaeus believed *Nomina si nescis, perit et cognition rerum*: “If you know not the names of things, the knowledge of the things themselves perishes.”<sup>33</sup>

Historians regard Linnaeus’s greatest contribution as his method of naming species. This method, called binomial nomenclature, gives each species a unique, two-word name consisting of the genus and species name. For example, humans are *Homo sapiens*, or “wise human,” a name that references our big brains.<sup>34</sup> Linnaeus classified 7300 species into 1098 genera.<sup>35</sup> Many local names gave way to a binomial latinized official name. He named his favorite flower after himself, *Linnaea borealis*. Linnaeus named so many plants and animals that he has been called “God’s Registrar.”<sup>36</sup> It is poetic justice that *Linnaeus* is a latinized form of *Linné*.

Endersby argues that the simplicity of the Linnaean sexual system helped it become popular, influential, and critical to the continued expansion of European knowledge of the world.<sup>37</sup> But his sexual system came at a price. Because of Linnaeus’s focus on the sexual anatomy of flowers, botanists such as Johann Georg Siegesbeck accused him of subjecting his students to immoral influences.<sup>38</sup> Other biologists believed that taxonomy ought to reveal more about the world than merely create distinctions.<sup>39</sup> Some challenged the curious and singular focus on the reproductive system; Linnaeus paid no heed to the many other plant features that could produce a more “natural” system. At one point, some scientists complained that despite decades of work, “nature’s order had still failed to reveal itself.”<sup>40</sup> Even for Linnaeus, such a revelation was the ultimate goal. In 1751 in the *Philosophia Botanica* he wrote: “The true beginning and end of botany is the natural system,” a system in which “all plants exhibit mutual affinities, as territories on a geographical map.”<sup>41</sup>

The goal of the natural system was to pay attention to the overall morphology of a plant, not just its flowers or fruits. The British generally remained enthusiastic users of the Linnaean system, making it popular in Britain and its colonies, histories particularly influential for botany. However, tensions remained between several classificatory systems. Today, the rules by which we name plants still depend largely on Linnaeus, although for plant families, A. L. de Jussieu’s classification in his *Genera Plantarum* of 1789 remains important.<sup>42</sup> By 1799, there were fifty-two different systems of botany, each taxonomy emphasizing a different part of a plant or plants’ geography.<sup>43</sup> Ultimately, by

the end of the eighteenth century, the Linnaean system beat out the others and emerged as the norm.<sup>44</sup> Despite the advent of evolutionary methods and DNA technologies, and despite its many critics and shortcomings, Linnaean classification remains remarkably resilient.

Empires, especially the Spanish, embraced the Linnaean system. Further, as King astutely argues, it seemed functional precisely because it relied on a social understanding of marriage and normative modes of sexuality.<sup>45</sup> Linnaeus's analogies between plants and humans, particularly about sexuality and the distinctions between male and female, made his work accessible.<sup>46</sup> He devised his *methodus propria* (proper method) as a temporary solution.<sup>47</sup> Only later in life did he realize that this temporary shorthand of a system would become his enduring legacy.<sup>48</sup> Linnaeus explicitly included rather than excluded women from the study of botany.<sup>49</sup> He reached out to lay and learned audiences, encouraged translations of his work, and in short, worked hard to democratize the accessibility and usefulness of his classification.<sup>50</sup> This broad dissemination and popularization were immensely successful. Unlike other classificatory systems, Linnaeus's caught on, creating the emergence of a "botanical vernacular" in literature and popular culture—perhaps the combined result of its relative simplicity as a taxonomical system and its vivid imagery of the sexual.<sup>51</sup>

The power of the Linnaean system is that it is both classificatory and nomenclatural.<sup>52</sup> Linnaeus published his classification system in the 1700s. Since then, many new species have been discovered. Initially, Linnaean ranks (kingdom, phylum or division, and so on) were thought to be objective because to him they were the plan of the creator, but with time, the Linnaean system was seen as not natural because it focused only on the sexual parts of plants, and thus needed revision.<sup>53</sup> Linnaeus's success was never absolute—for example, the Jussieuan system found favor among professional proponents in the 1830s—but Linnaeus endured in the "botanical vernacular." Thus, King argues that while the line between professional and amateur natural history hardened in the first half of the nineteenth century, Linnaeus retained a strong presence in the vernacular but not the professional realm of botany.<sup>54</sup>

**COLONIAL ROOTS AND AFTERLIVES** The power of naming in Christian cultures is a divine power. In Genesis, Adam is given the power to name the animals. In the botanical realm, botanists claimed a "natural" and God-

given power to name.<sup>55</sup> This power emerges in the language and practices of imperialism. Christopher Columbus named the first American island he encountered “*San Salvador*, in remembrance of the Divine Majesty, Who marvelously bestowed all this.”<sup>56</sup> Not only did Columbus take possession on behalf of God and the Spanish monarchy, but he claimed a righteous power to do it even while knowing that the local Indians called the island “*guanahani*.” In asserting his power, he erased native nomenclature.<sup>57</sup> This colonial practice of naming is at the heart of scientific epistemology—naming places a barrier between observer and observed, a separation between scientist and nature. Here, the scientist is all supreme.<sup>58</sup> Latinized scientific names, unlike local names based on plant characteristics, render plants alien. This attitude is pervasive; organisms were named by the colonizers to celebrate European legacies—God, Queen, King, plant “discoverer,” friend. Indeed, the very term *animal* (or, for that matter, *plant*), bundling together disparate creatures despite their many differences, speaks to this power.<sup>59</sup> As Prosek reflects, “I think what bothered me was his hubris. . . . Plants and animals already had names in indigenous languages, and Linnaeus, in a show of imperialism, renamed them with his Latinisms. He believed he could take nature—holistic, fluid and constantly changing—and fragment, label and systematize it.”<sup>60</sup>

Linnaeus was a superb promoter of his work. Indeed, the global exchange of seeds and plants created a vibrant political economy for Linnaean botany.<sup>61</sup> He was deeply engaged in global botanical exchanges and “trade relations,” establishing himself as “the most consummate botanist.”<sup>62</sup> Lisbeth Kroerner’s excellent biography of Linnaeus describes his expedition to Lapland, the northernmost region in Sweden, between May and October in 1732 as an important part of his scientific career, especially in how he positions himself vis-à-vis the indigenous.<sup>63</sup> By the 1600s, a century before Linnaeus ventured forth, the indigenous Sami had already lost their native religion and been converted to Christianity by missionaries; they had become “tourist guides of their own culture’s destruction.”<sup>64</sup> Given this, it is particularly ironic how Linnaeus exaggerates the hardships he encountered and overestimates the miles he traveled (he was being paid by the mile). The Sami emerge in Linnaeus’s *Flora Lapponica* as primitive and untouched, and they and their lands are portrayed as alien and strange. Linnaeus writes: “It seemed as if I were entering a new world; and as I climbed higher I did not know whether I was in Asia or Africa, for the soil,

the situation and all the plants were strange to me. . . . There were so many that I feared I was taking away more than I would be able to deal with.”<sup>65</sup>

His description and claims about the indigenous Sami are particularly illustrative. Linnaeus presented his Lapland travels as “an empirical enterprise and an adventure upon a *tabula rasa*.” He dismissed older texts and remade the Sami as savage modeled on “international ethnography on indigenes of America, India, and Western Africa as well as on Ossentians, Samoyeds, Inuits and Kalyks.” Yet he spent only eighteen days of his five-month journey outside the homes of Swedish homesteaders or on the coast. Koerner notes,

Linnaeus mythologized his sub-Artic travels into a formative encounter with an Edenic “wild nation” (Sami reindeer herders) and a cross-cultural encounter between “high” and “folk” science. He then used this performative narrative to enter Dutch learned circles. Yet the Lapland journey was also part of the colonization of “our West Indies,” as Scandinavians termed their Arctic frontier. And this colonial venture in turn was predicated on erasing indigenous culture, as the “wild” Sami and their herds were chained to the engines of cameralist industry.<sup>66</sup>

Geoff Bil extends this insight to argue that the act of plant classification was profoundly a naturecultural project. European fieldworkers saw local plant names as a potentially “lucrative storehouse of indigenous knowledge.”<sup>67</sup> Beyond botanical fieldwork, indigenous plant names opened a window into producing colonial ideas of indigenous cultural difference. Bil chronicles the rich and complex histories of plant biology and ethnology where indigenous phytonyms become the basis of claims of racial and cognitive differences of local peoples. For example, Alexander Humboldt claimed an inverse correlation between plant diversity and cultural sophistication. European societies, he preposterously argued, compensated for their impoverished botanical heritage with richer languages and intellects. Thus, Linnaean plant names helped produce stable, mobile, and universal systems, critical to the enterprise of mapping “imagined and actual worlds” of plants and people.

The Eurocentric and colonial disposition of Linnaeus endure. The nomenclature adopted by the Vienna 1905 International Congress of Botany echoes the colonial legacies of Linnaeus. Consider the following:

V. Botanists who are publishing generic names show judgement and taste by attending to the following recommendations: . . .

- c) Not to dedicate genera to persons who are in all respects strangers to botany, or at least to natural science, nor to persons quite unknown.
- d) Not to take names from barbarous tongues, unless those names are frequently quoted in books of travel, and have an agreeable form that is readily adapted to the Latin tongue and to the tongues of civilized countries.<sup>68</sup>

While colonial ambitions were many, colonialism was often rationalized as a “civilizational” mission of lesser humans.<sup>69</sup> The above criteria—with terms such as *judgement*, *taste*, *persons strange to botany*, *barbarous tongues*, and *civilized countries*—are barely disguised code for colonially professed “superior” European norms distinct from those of indigenous and colonized peoples. By this logic, the barbarous, non-Latin-tongued people could be lovers of plants, in their primitive ways, but not the purveyors of botany.

#### THE RISE OF EVOLUTIONARY SCIENCES AND THE ADVENT OF DNA

Today, the impulse to name organisms or create more accurate “natural” classificatory systems of plants is no longer a task to uncover divine order (as for Linnaeus) or for the ease of living. Rather, the overarching goal of systematic biology is to produce a comprehensive and accessible catalog of life on Earth.<sup>70</sup> While traditional taxonomy encompasses the description, identification, nomenclature, and classification of organisms, the subfield of systematics is dedicated to evolutionary relationships between organisms. The rise of evolutionary sciences, and the advent of modern systematics, reframed biological classification as the evolutionary relationships of taxa. Phylogenetic descent or evolutionary classification became the grounds for taxonomy, and genetic information replaced morphology as the critical site of data. The search for a “natural system” is ongoing.

With technological advances in the 1950s, molecular biologists found confidence bordering on imperial zeal, creating a rift with evolutionary biology.<sup>71</sup> Evolutionary biologists believed that natural selection worked at the level of the organism, while molecular biologists insisted that genetic data were key. Molecular biologists argued that mutations in the genetic code were the key source of variation; evolutionary biologists, on the other hand, argued that natural selection, not DNA, acted primarily on the organism. Scientists in the fields upheld the superiority of their chosen field, while some tried to find a compromise where the two fields would complement each other.<sup>72</sup> In a wonderful summary, two molecular biologists argued that DNA is not a passive carrier of the evolutionary message: “Evolutionary change is not imposed upon DNA from without; it arises from within. Natural selection is the editor, rather than the composer, of the genetic message.”<sup>73</sup>

In the mid-1980s computational power transformed biology into a data-driven science. The wet labs (full of lab benches) were transformed into dry labs (full of computers).<sup>74</sup> While wet labs and their work continue, hardware and software are critical in an increasingly data-driven field. Systematics, a field founded on the principles of evolution and a goal to “discover” the most accurate phylogeny of life on earth, moved to using vast data sets to ascertain evolutionary relationships.<sup>75</sup> It is assumed that the more characteristics one includes, the greater the accuracy. With these innovations, intense disagreements emerged and continue. As Bill Bryson quips, “Taxonomy is described sometimes as a science and sometimes as an art, but really it’s a battleground.”<sup>76</sup>

**THE TREE OF LIFE** The tree of life emerged as the foundation and emblem for the biological sciences, much like the periodic table in chemistry. While systematists argue about how best to construct phylogenetic trees, there is still a deep impulse for “tree thinking.”<sup>77</sup> Phylogeny (a term coined by biologist Ernst Haeckel), or the generation of species, implies that closely related species share more traits with one another. Once species diverge, their evolution takes them along different branches to new adaptations and innovations. As Darwin wrote to a friend, “The time will come I believe, though I shall not live to see it, when we shall have fairly true genealogical trees of each great kingdom of nature.”<sup>78</sup> This is always the quest of western science: a singular, universal, and unified truth.

But on what basis does one organize an evolutionary tree? What traits and changes mark evolutionary change? Depending on which traits or combination of traits one selects, the phylogenetic trees that emerge can be wildly different. The field moved from typological thinking, where organisms and groups had “essential natures” and where variation was largely ignored, to a model focused on populations and variation. These ideas—population and variation—are immensely vague and politically malleable categories.

In modern systematics, the most influential figure is the German biologist Willi Hennig, who founded the field of cladistics.<sup>79</sup> Hennig organized groups that shared a common ancestor through the structure of a branching tree.<sup>80</sup> Cladistics is organized into monophyletic groups: the most recent common ancestor and all its descendants. At the base are ancestral organisms that branch (or not) into the tips with extant species. In cladistics terminology, all species have some characteristics that are ancestral (plesiomorphic) or derived (apomorphic) relative to other species. For example, for reptiles, scales and four limbs are ancestral traits. Lizards have both scales and four limbs, whereas snakes have scales (ancestral) but no limbs (derived). This does not imply that one is superior to the other; phylogenetic trees merely capture the acquisition and loss of characteristics, which can happen multiple times. The key point here is that we should not talk about any species that lives today as primitive, ancestral, lower, advanced, or higher. All extant taxa are of the same age, having evolved for the same amount of time from a common ancestor some 3 or 4 billion years old. The tree-thinking view of species is thus quite egalitarian—it organizes life, past and present—and helps us visualize varied evolutionary adaptations and variation but ultimately celebrates all species.

With the advent of cladistics and, more recently, computing power and molecular technologies, evolutionary genetics and the genetic sciences have taken over the field. Genetic data and the discovery of many genes in plants have produced data for new evolutionary trees. But not all scientists are converts. Some taxonomists still work with morphological characters using the power of large databases, a synthesis not possible earlier. Others, unconvinced that genes are the primary site of evolution, continue to ground evolutionary thinking in the adaptations of organisms.

## Infrastructures of Naming, Nomenclatures, and Classifications

In their exploration of classification and its importance, Bowker and Starr argue that classifications and standards occupy a critical place in the studies of social order.<sup>81</sup> Modern systematics might reclassify and reshape our understanding of evolution, but much of plant sciences outside academia have practical purposes—agriculture, horticulture, and the worlds of nature lovers and amateur naturalists. Academics and their focus on phylogenetic relationships don't always see eye to eye with applied botanists outside of academia.

In working on this history, I have come to appreciate the vast (yet invisible to most) human infrastructures and standards that keep the botanical world functioning. Embedded in histories of infrastructure are organizational histories of taxonomists and their naming protocols.<sup>82</sup> Classification and standards are closely related but not identical. Over the history of botany, many have proposed classificatory systems, but as we know, most did not survive or become the standard. In some instances, multiple naming systems coexist.

How are nomenclatures regulated? I want to briefly explore the very subjective and deeply Eurocentric roots of nomenclatures that standardize botany. Critically, I want to explore what standardization does, whose standards prevail, whose knowledge is erased, and what is lost. I want to ask how botanical worlds might have been otherwise and how we might access these histories for the future.

Botanical nomenclature has never been fully aligned with botanical classification. Decisions on nomenclatures are not arbitrary but are regulated by the taxonomic community through a “family planning” committee that meets regularly.<sup>83</sup> In 1982, for example, International Botanical Congress in London adopted a set of basic rules. To quote Gledhill, they included the following:

1. One plant species shall have no more than one name.
2. No two plant species shall share the same name.
3. If a plant has two names, the name which is valid shall be that which was the earliest one to be published after 1753.
4. The author's name shall be cited, after the name of the plant, in order

to establish the sense in which the name is used and its priority over other names.<sup>84</sup>

Rules such as these are considered at each International Botanical Congress held every five or six years.<sup>85</sup> The two-century-old effort to establish and stabilize scientific naming of plants has not been easy. In recent years scholars from formerly colonized nations have challenged this system. A lot is at stake. Two examples give a flavor of the debates.

Some systematists in 1999 proposed that the tomato should move from the genus *Lycopersicon* to the genus *Solanum*.<sup>86</sup> While this change may satisfy academics, for seed companies, the costs of relabeling, repackaging, and advertising could run in tens of millions of dollars. A debate ensued. Ultimately, the tomato was renamed from *L. esculentum* to *S. lycopersicum*. If this happens repeatedly, costs can be considerable. For the buyer and seller of tomatoes, does the scientific nomenclature matter? Should it?

More recent is the case of the genus *Acacia*, a massive cosmopolitan genus across Africa, Asia, the Americas, and Australia. Growing genetic evidence suggested that the genus was not monophyletic (that is, descended from a common evolutionary ancestor). Some proposed a split: one lineage in Australia and another in South Africa. But which would retain the name *Acacia*? Which local species would emerge as the type species? Nature and nationalism aligned—both countries claimed the acacias. A passionate and contentious debate ensued with strong political and nationalist fervor. Ultimately, the 2005 International Botanical Congress in Vienna approved a decision to conserve the name *Acacia* for Australia while two pan-tropical lineages were named *Vachellia* and *Senegalia*. Interestingly, this was possible because the committee adopted minority rule even while the majority voted against this decision. Arguing for a procedural violation, this issue was revisited in 2011 but with similar results.<sup>87</sup>

Many professional and amateur botanists in Africa, Asia, and Central America were incensed by the decision, arguing that a well-financed, politicized campaign by Australia was behind it.<sup>88</sup> It ignored the tree's African symbolism. Does retyping the genus signify "yet another act of economics and cultural pillage of the Third World by the First?"<sup>89</sup> The website for the

South African National Parks, under a banner headline “Africa to lose all its Acacias,” pointed out that nine people had decided for the entire world. A newspaper in Nairobi expressed outrage in the headline, “Did you know it is illegal to call this tree Acacia? Australia claims exclusive rights to the name.”<sup>90</sup> So deep is the opposition that despite the decision, several authors writing about African acacias have pointedly retained the genus name *Acacia*.

This split between academic botany and practitioners in the field repeatedly emerges. For example, gardeners find broad categories of roses, cabbages, carnations, and leeks to be perfectly adequate for their purposes. They are satisfied with a naming system that does not keep up with botanical rules or reassignments based on molecular information. Indeed, many seed and plant catalogs avoid botanical names altogether. Changing names as per cladistics and systematics is annoying and costly. However, if names are not changed when new information is discovered, then naming loses its connection with theory and emerges as an arbitrary mapping of world without any system governing it.<sup>91</sup>

Some worry that an exclusive focus on academic evidence in the name of scientific objectivity denies the feelings for local traditions in plant names, undermining the ability of the *International Code of Botanical Nomenclature* to serve as a universal system.<sup>92</sup> There is value in local knowledge. Sometimes it is more insightful and long lasting. For example, oral traditions in Hawai‘i have accurately identified individual algal species, while three-fourths of the scientific names have changed in the past ninety years.<sup>93</sup>

## Botany as an Imperial Archive

A central impulse in the history of taxonomy is creating an imperial archive. During colonial times, taxonomy helped consolidate knowledge for a more efficient extractive empire. Today, its goal is to chronicle all life on earth. Biologist E. O. Wilson writes:

Now it is time to expand laterally to get on with the great Linnaean enterprise and finish mapping the biosphere . . . discover and describe the Earth’s species, to complete the framework of classification around which biology is organized, and to use information technology to make this knowledge available around the world.<sup>94</sup>

Bowker characterizes such a view as a “panoptical” dream, an enduring impulse toward an imperial archive.<sup>95</sup> Indeed, many have attempted such inventories.<sup>96</sup> A few examples include the Barcode of Life (BoL), a project literally inspired by supermarket barcodes, and a crowd-sourced inventory by systematists and qualified amateurs to produce a Wikipedia-styled website, Encyclopedia of Life (EoL).<sup>97</sup> Earth BioGenome Project, a multibillion-dollar venture, attempts to catalog the genomes of all species on earth and is labeled by its organizers as a “moonshot for biology.” The ensuing sequences will be used to create a Digital Library of Life (Nelson 2018). Likewise, the E. O. Wilson’s Biodiversity Foundation’s Half-Earth Project aims to create an Inventory of Life.<sup>98</sup> The move to register all names, to agree on model data structures and formats for botanical databases to facilitate biodiversity management, is just as urgent and just as overly optimistic as Alphonse de Candolle’s calls in the nineteenth century for a rational system of nomenclature.<sup>99</sup> Such incomplete utopian projects are so pervasive in the history of naming and record-keeping that they should be regarded as standard rather than abnormal.<sup>100</sup>

Despite such initiatives, taxonomists lament the decline of the field, at a time when the “sixth extinction” is upon us.<sup>101</sup> We are losing species much faster than we can discover and name them,<sup>102</sup> and while taxonomists tout abstract notions of chronicling biological, in reality there is inadequate capacity for this. The field remains Eurocentric, with little organizational interface to facilitate a fair exchange of information.<sup>103</sup> There continues to be a “*de facto* information imperialism, causing a net flow of raw data out of the Third World into Western databanks—where it is converted into economically valuable information and knowledge, and then sold back.”<sup>104</sup> Like in colonial times, western scientists continue to “parachute” into third world nations, access their botanical resources, and return home to process their data.<sup>105</sup> It is an old problem—third-world countries catching up with first-world technology, and poor community groups bridging the great divide.<sup>106</sup>

The project of decolonization necessitates deep reflection of such ongoing colonial practices. How do we build collaborations between scientists and institutions in formerly colonized worlds? How can international infrastructures undo colonial hierarchies of power? The answers are decidedly complex and rarely satisfying or equitable. For example, although indigenous peoples have used certain plants for centuries, patents still elude them within west-

ern legal regimes. At best, it is possible to prevent individual scientists and companies from being granted exclusive patents. Fascinating cases involve indigenous groups, growers, scientists, and multinational companies working to craft workable contracts and profit-sharing agreements. In *Bitter Roots*, Abena Dove Osseo-Asare tells the story of six African plants and the production of pharmaceutical products from them.<sup>107</sup> The indigenous San peoples have long used Hoodia to reduce hunger, increase energy, and ease breastfeeding. In *Reinventing Hoodia*, Laura Foster chronicles how the plant was reinvented through law, science, and the marketplace and collaboratively repackaged as an appetite suppressant.<sup>108</sup> Indigenous peoples, scientists, growers, and multinational companies negotiate contractual terms within uneven political and legal landscapes. While indigenous groups have pushed back to create more equitable arrangements, colonial botany has laid the foundations for its enduring power.

Plant “types,” the original specimens used to typify a species, are essential references during identification.<sup>109</sup> As we have seen, types are not always hosted in the country where they were collected and are usually held in western nations. For example, in their analysis of the family *Rubiaceae* in Angola, they found that in 430 taxa (96 percent), the actual type was in European herbaria. Only 50 taxa (12 percent) have type material represented in African herbaria. Colonial legacies endure in their afterlives. Yet there is a deep reluctance to change or undo these legacies.<sup>110</sup>

In the impulse to decolonize, should materials be repatriated? Decision makers and politicians prefer repatriating digital images and databases rather than the physical archive.<sup>111</sup> Ironically, at the same time we have seen a disinvestment in western archives and a decline in the number of taxonomists working in those archives. Young African taxonomists are frustrated about not being able to access, let alone maintain and expand, their work while European institutions grow moribund without functioning budgets or staff, “where valuable African specimens are deposited, uncured and unused, running the risk of slow, inevitable decline, if not decomposition.”<sup>112</sup> In short, postcolonial nations that have invested in expertise have no access to control the representations of flora of their own nations. An exception is an initiative, Flora of Ethiopia, where external research funding has worked alongside building of local infrastructures.<sup>113</sup>

The Rhodes Must Fall campaign, which started in Cape Town, South Africa, in 2015 and then spread across the world, has added urgency to these questions. Cecil John Rhodes, a colonist, imperialist, and racist and “a symbol of the ruthless imperialism,” has 126 plants (collected between 1899 and 1976) that bear his name. The name of Rhodes has been erased from the names of nations (Rhodesia), and the recent South African campaign challenged the presence of his legacy in the statues and in the names of institutions. Yet some activists astutely ask: why is he still celebrated in the botanical world?<sup>114</sup> Rhodes is, of course, only one of many colonists who continue to be honored in the botanical imagination. During colonial rule, white colonists replaced local names with those of royal figures and the names of white botanists. Why should those names remain? Why should botany reinforce colonialism through a parade of western names? With the rise of digital media, change should be easy.

Many environmental organizations such as Sierra Club and the Audubon Society are reckoning with the nativist and racist beliefs of their founders. The continued link of environmental groups to nativist and xenophobic ideas, dubbed the “greening of hate,” is a reminder of enduring racist and colonial legacies.<sup>115</sup> While the national chapter of the Audubon Society has retained its founder’s name, some local chapters have opted for new names.<sup>116</sup> The American Ornithological Society announced on November 1, 2023, that it would change the names of all bird species named after people. Further, it committed to involve a diverse representation of experts and the public in the process.<sup>117</sup>

Questions of control of botanical infrastructures also pose a problem. For example, botanical repositories in the planet’s tropical, subtropical, and temperate regions house a wealth of global flora. Yet, as we have seen in the case of *Acacia*, those countries hold only a fraction of the votes necessary to amend the *International Code of Botanical Nomenclature*. Colonial-era rules have proved difficult to change given the enduring coloniality of power.<sup>118</sup> Today, plant names, nomenclatures, and taxonomies are all abstract, scientific exercises. No doubt they are useful and important to the life sciences, but they contain less utility for the nonspecialist plant lover. One of the problems with the taxonomic system is that it is not designed to be an information retrieval device. In folk taxonomies, names are practical and pragmatic—they provide contextual information to others who already share an understanding of the organism’s cultural significance.<sup>119</sup>

It is important to note that other imaginations in botanical history offer alternatives. For example, Sir William Jones, an oriental scholar and linguist, developed a great interest in botany. As a philologist, he used indigenous knowledges and languages.<sup>120</sup> He drew often on Sanskrit names for plants, arguing, “I am very solicitous to give Indian plants their true *Indian* appellations.”<sup>121</sup> He gave plants a Sanskrit name, a uninomial, followed beneath by its common name, and beneath these its Linnaean or other scientific equivalent. In part the local Indian languages were provided to make resource identification and extraction more efficient.<sup>122</sup> Alas, he published only two papers on Indian plants, both posthumously.<sup>123</sup>

Plant sciences focus on organisms that bring funding. Not all organisms receive equal attention, even commensurate with their importance. For example, there are more viruses than bacteria in the open ocean.<sup>124</sup> But viruses, though plentiful, do not get as much attention as bacteria. Similarly, there is a bias toward flowering plants, especially those that are big, exotic looking, tasty, or economically important. Many of the most urbanized coastal areas where human impact has been greatest have never been surveyed.<sup>125</sup> An analysis of plant protection legislation in twenty-nine European countries lists taxa almost entirely composed of vascular plants and dominated by spectacular species attractive to collectors and the public.<sup>126</sup> Despite claims of “objective knowledge,” western botany retains its fascination with the exotic other.<sup>127</sup>

## THE COLONIALITY OF POWER

Sylvia Wynter, drawing on Aníbal Quijano, Walter Dignolo, and Michel Foucault among others, offers a foundational story of colonialism.<sup>128</sup> Colonial knowledge systems, she argues, transformed the human into Man. The vast intellectual enterprise of western civilization of “Man” should not be confused with the human or humanity. To unsettle the coloniality of power, she argues, it is critical to unsettle the overrepresentation of Man, a central secular figure of modernity. Wynter identifies two historical stages—the first from the Renaissance to the eighteenth century, when Man was secularized into a rational political subject (Man1, *Homo religiousus* or *Homo politicus*), and the second from the eighteenth century until today, which captures the Darwinian vision of the naturally selected Europeans (Man2, *Homo oeconomicus*). *Homo reli-*

*gioursus* emerged from a “theological order of knowledge of pre-Renaissance Latin-Christian Medieval Europe,” a figure that ushered in *homo oeconomicus*, a bourgeois figure of liberal mono-humanism from the late nineteenth century. Man1 makes possible the development of the physical sciences, and Man2 the rise of the biological sciences.<sup>129</sup>

Man2 is the figure of the colonial botanist. To understand why this is important, Wynter narrates theology—the Christian God was reconceived as a caring father who created the world/universe for mankind’s sake. This made way for Copernicus’s declaration that since the universe was made for our sake, it was knowable. This belief undergirds the power of botany, plant naming. The shift from divine providence and/or retribution, and from witchcraft and sorcery, to the new principles and laws of nature ushered in modern science and medicine. For the history of botany, it is important to recognize that “nature” became the “representative agent” of God on earth. This “nature” is the object of scientific inquiry.

I spend time discussing coloniality for several reasons. First, it helps us understand how colonialism is at the root of modern science and its rise. Second, it helps explain why and how the western human created hierarchies of being—racial and gender hierarchies in humans and classification hierarchies among the nonhuman. Plants, seen as nonsentient and immobile, were at the bottom of such hierarchies and thus were turned into objects of science and commerce. Among humans, European men were considered superior, followed by white women and other humans (if indeed they were considered human at all). Third, it’s important to understand coloniality because the sciences birthed from colonialism inherited many of its characters—an abstract “natural” world for human consumption, the idea of a disembodied scientist, the practices of an objective scientific culture, and finally the supremacy of science and the scientist as the site of knowledge. Finally, colonialism and colonial ideology knit the world together through academic knowledge. While we inhabit disciplinary silos today, they are connected historically. The history of botany is not a singular history of plants but a colonial worldview and ideology imposed upon all who were not the figure of “Man.” Any project of decolonizing botany must simultaneously also decolonize a worldview and ideology. The history of botany is a story about more than plant worlds.

I began this chapter discussing how naming and organizing living beings were meant to ensure *order* in the world. Theories of evolution and DNA technologies have transformed how we understand life on earth. The idea of the tree remains central to the evolutionary imagination, a motif for the branching and deeply rooted connectedness of all life on earth. At the end of the branches are discrete species of life. In *Imperial Leather*, Anne McClintock powerfully unpacks two central genealogies of imperial nature: the combined figures of the tree of life and the white “family of man.”<sup>130</sup> Central to this book is how gender and race traverse species lines to ground hierarchies of difference of both the social and biological, the human and nonhuman alike. In short, the story of “man,” where hierarchies of race and gender are saturated with meaning, is also the story of the nonhuman. Colonial and repressive actions often appeal to both ideas of nature and the family.

By now rich accounts of colonial botany have persuasively argued that imperialism has profoundly shaped our view of nature—how we know nature, our relationship to nature, the individuals who produce knowledge, and our theories of the natural world.<sup>131</sup>

The figure of the tree has transformed as new technologies have arisen. Human phylogenetic trees visualize kinship not between species but between human “populations.” The tree of life’s representation of evolution has shifted from an earlier hierarchical model of evolution as a ladder to a branching structure of partition and diversification, although trees as a motif tend to downplay genetic variation within species. One limiting factor of the tree is that the branches can never merge. Here, histories of biology inspire. Julian Huxley, challenging racialized views within biology in 1936, argued that unlike plant and animal species, there was too much interbreeding throughout human history for speciation to occur or for there to exist isolated and unique populations.<sup>132</sup> Instead of a branching tree, Huxley introduced the idea of reticulate evolution where branches merge and join each other—a network rather than a tree.

The diverging tree is embedded in evolutionary and popular imagination with a singular focus on biological difference, most prominently gender and racial differences.<sup>133</sup> The Human Genome Diversity Project as started in the

1990s with the idea that genetics could help us understand human migration and evolution. Jenny Reardon powerfully shows how the World Council of Indigenous Peoples saw this approach as fundamentally racist and colonial—a “vampire project.”<sup>134</sup> As Reardon argues, population genetic studies, like all scientific studies, are never “pure” but always a coproduction of the social and natural order. Similarly, Lisa Gannett reminds us that ideas about the biological concept of “race,” while ostensibly rejected after World War II, never disappeared but were recoded in the terminology of “population” thinking. Powerful colonial ideas and tropes—like gender and race—have afterlives because they became naturalized as scientific concepts. Race gets increasingly codified within evolutionary language rather than ecological ones, perpetuating the language of fundamental genetic differences among races.<sup>135</sup> In this view, we are not our environment, we are our genes. These tensions and representations of evolution also shape our view of plants. Given this racialized history, should we rethink the very idea of species?<sup>136</sup>

Huxley’s proposal of reticulated evolution has reemerged in recent years as a model for life on earth. Interbreeding between one branch and another can add layers of complexity. In addition, works by Lynn Margulis, Tsutomu Watanabe, Carl Woese, and W. F. Doolittle have challenged the exclusive focus on vertical inheritance across generations and instead highlighted the lateral movement of genetic material across species lines—that is, horizontal gene transfer.<sup>137</sup> Barbara McClintock’s work has also highlighted the movement of transposable elements, a process that challenges simplistic Mendelian inheritance. Roving genes have now been found in every branch of the tree of life where geneticists have looked: “It’s not just rare freaks or accidents, it’s happening all the time. And in quite divergent species too.”<sup>138</sup> This movement of DNA across species and across time fundamentally challenges any neat phylogenetic tree and also the idea of biological individuality.<sup>139</sup> If organisms transfer and accept DNA across species within a generation, how unique is the individual? In the field of human paleoanthropology, some scientists have suggested a metaphor of a braided stream, where streams can branch out but also flow back into the main river.<sup>140</sup> Recent work suggests that we should move away from the tree to a network or reticulated evolution.<sup>141</sup>

I began with the messy world of postcolonial naming—naming and renaming, multiple namings, sedimented meanings. Models of reticulated evolution

in a profoundly naturecultural world inspire the title of this chapter as the “categorical impurative.” Throughout botany’s history, scientists have attempted to come up with a definitive genealogical tree of life, a goal that proves elusive. Emily Singer argues that “the current version of the tree of life is more like a contentious wiki page than a published book, with certain branches subject to frequent debate.”<sup>142</sup> Indeed, one geneticist, after a study of yeast populations, concluded, “We are trying to figure out the phylogenetic relationships of 1.8 million species and can’t even sort out 20 types of yeast!”<sup>143</sup> As the history of nomenclature and classification has taught us, the problem is not the lack of the right technique or biological variable or theory. The problem is the imperial quest for a universal, unitary, pure genealogy of life on earth. Perhaps there is no unitary order? Perhaps we should put quests of purity aside to embrace the ebullience of plant adaptations, the diversity of wondrous morphologies, and the messiness of life on earth. Embrace the categorical impurative!